

Game Based Learning Model for History Courseware: A Preliminary Analysis

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Abstract

History learning is vital for students' intellectual, spiritual, emotional and physical development. Previous researches showed that there are problems in history learning among students. Students perceive history subject as difficult and very boring because they have to memorize facts besides understanding those facts, concepts, time and historical events. Interactive multimedia can increase students' enthusiasm and make the teaching and learning process more interesting and make the history 'alive'. Digital games, an interactive technology could foster learning process effectively and interestingly especially among young learners. In this paper, we present findings from a preliminary analysis on problems in history teaching and learning, students' readiness for digital game-based learning and the capabilities of digital games for history learning. We did a document analysis of SPM performance report from year 2002 to 2005 and further investigation was carried out using questionnaire and interview, involving 582 form four students and fifteen history teachers from five secondary schools in Kajang, Selangor. The key findings from analysis are types of problems faced by the students in learning history and the potential of Game Based Learning (GBL) approach for the development of interactive courseware to increase students' interest in learning history via an entertained experience. Therefore, we propose a GBL Model for the design of history courseware.

1. Introduction

History subject plays a vital role in instilling the spirits of patriotism among the students; to nourish and instill love and loyalty to one's country and to be a true citizen [6, 32]. Through knowledge and appreciation of our country's history and other country's history, students have some understanding of his/her society

and country and the relationship of the country to the world's history. Furthermore, teaching and learning of history is appropriate for students' intellectual, spiritual, emotional and physical development as prescribed by the National Education Philosophy [13, 21]. Nevertheless, there are problems in the teaching and learning of history.

Rapid development of ICT has lead to widespread and effective usage of the technology for teaching and learning since ICT can increase students' enthusiasm and make teaching and learning process more interesting, beside making history 'alive' [1]. Electronic games are growing rapidly as a cultural form, a set of media technologies and a global industry [22]. Computer and video games have recently attracted significant attention from educators as a medium for learning [34].

Digital games, an interactive piece of technology within the multimedia learning environment could foster the learning process effectively and interestingly especially among young learners. Researchers and game designers have noted this promising technology and proposed some frameworks and models to foster multimedia learning environment [36]. Game based learning (GBL) is the subject of a huge amount of academic research [5].

In this paper we present findings from a preliminary analysis of problems in history learning at secondary school level and students' readiness for digital game-based learning and the capabilities of digital games for history learning. We then propose a GBL Model for history courseware design.

2. Digital games

Game is defined as a physical or mental contest with a goal or objective, played according to a framework or rules which determined what a player can and can not do inside a game world [2]. Digital games is defined as games which provided visual information to one or more players, accepted input from the players

and use a set of programmed rules [26]. Digital games are virtual play environments with features such as rules, goals, story, interaction, systems series, Nintendo GameCube™, Microsoft's Xbox® and Xbox 360™, games-specific handheld consoles (example: Nintendo's Game Boy® and DS™ series, Sony's PSP™), DVD-ROMs or CD-ROMs on personal computers either connected to a network or disconnected or handheld units (example: personal digital assistants, portable computers or mobile phones) [14]. Action games, adventures games, fighting games, puzzle games, role playing games, sports games, puzzle games and strategy games are the genres of the digital games [14, 17, 26].

Games are played primarily for entertainment [2]. But nowadays games have been integrated into education to form an innovative educational paradigm [36]. Besides, games are also an extremely valuable context for the study of cognition within this paradigm [34]. The educational potential of computer games are giving learners a "mental workout" since the structure of activities embedded in computer games can develop a number of cognitive skills [29].

3. Game Based Learning

Game Based Learning (GBL) is a paradigm which utilizes the game as a medium for conveying the learning contents [36]. GBL is all about leveraging the power of computer games to captivate and engage end users for a specific purpose, such as to develop new knowledge and skills [5]. GBL is also defined as an application which uses the characteristics of video and computer games to create engaging and immersive learning experiences for delivering specified learning goals, outcomes and experiences [10].

GBL is widely adopted in manufacturing, energy/utilities, computer technology, higher education, retail, financial, telecommunications, pharmaceutical, hospitality, construction, computer software, public departments and healthcare sectors [5].

From the past studies reviewed, the minimum components required in all online games for learning are back story and story line, games mechanics, rules, immersive graphical environment, interactivity, challenge/competition and risks [8]. However, according to Tan et al. [36], the two major components in GBL Models are (1) Pedagogical (difficult to learn, psychological needs, critical thinking, exploration, challenge, engagement, competition, practice, goal setting and motivation) and (2) Games Design (interaction, storytelling, interface, simulation, construction, feedback, literacy, communication, motor skill, memory and outcome evaluation). According to Clark [4], the main elements in GBL are

Design Specification (Story, Consequence, Rules and Interactivity) and Engagement (Emotional, Psychomotor and Intellectual).

According to Garzotto [11], four major independent factors that contribute to the educational effectiveness of a multiplayer e-gaming experience are usability (ease of use, interface), content (target/goal appropriateness, integration, scaffolding, extensibility, media matching.), enjoyment (clear goals, concentration, challenge, feedback and immersion) and social interaction (connection, cooperation, competition).

4. Teaching and learning of history

This section discusses the problems in teaching and learning of history, the GBL approach to teaching and learning of history and the proposed components in GBL Model for history courseware design. We also present results of preliminary analysis carried out at local secondary schools.

4.1 Problem of learning History subject

According to Azwan et al. [1], Curriculum Development Division reported that history subject is known as a 'dead' and boring subject. In addition, our society presume that history subject do not have commercial value. Furthermore, the problems faced by history teachers is that student have no interest in studying history subject.

Students' impression of history subject is that it is difficult and very boring because one has to memorize all facts in the text book besides understand those facts, concepts, time and historical events [13, 32]. These students' attitude has indirectly caused the decrease in passing percentage for Malaysian Certificate of Education (SPM) examination and Higher School Certificate of Malaysia (STPM) [32]. Failing percentage of history subject for SPM examination in 2005 SPM was 25.6% [19].

This failing percentage increased to 34.3% in 2006 [3, 19, 37]. Failing percentage for history subject in 2005 STPM was 45.11% [18]. History was one of the subjects showing a decrease of more than 3% in passing percentage in SPM, thus necessitate a detailed analysis and study for this poor performance [23]. Moreover, a study by Rozita and Zaliza [32] found problems with history learning such as students' attitude; namely lazy to read, read but did not memorize facts, not interested in history subject, student interested but have to memorize too many facts, cannot manage to master the themes and the teaching and learning which is not effective. The lack of creativity in history teaching caused students to become bored and lose interest in the subject [1].

History teaching in school is less challenging for students' mind and intellectual development and learning by rote memorization also does not encourage students' analytical and critical thinking [13].

4.2 Findings from Preliminary Analysis

A preliminary analysis was carried out to identify problems and to determine the requirements to overcome these problems. We did a document analysis of SPM performance report from year 2002 to 2005 [19]. The report indicated that students in moderate performance group are able to give good historical facts but did not give suitable examples, and/or gave long irrelevant elaboration. Students in low performance group commented the questions without giving answers and repeated the same facts, answered one/two questions, did not understand questions and weak in language style. As for structured questions, students are confused, thus mixing all historical facts. They lacked understanding and do not use historical facts properly. Performance report (2002-2003) indicated that overall students' performance is less satisfactory. The report (2005) also indicated that students still did not answer questions properly but gave abstract answers.

Further investigation was carried out using questionnaire and interview. Respondents for this study are 582 form four students and fifteen teachers from five local secondary schools in Selangor. Results show that 79% of students have problems when learning history. Difficulty in memorizing facts is the main problem faced by 69% of the students. Teaching media such as boring text books also cause students to lose interest and thus face problems when learning history. Table 4.1 shows details of problems faced by students. Problems faced by students from teachers' perception (Table 4.2) also include difficulty in memorizing facts, besides having no interest in history. This finding is supported with data from the interview of form four history teachers [12, 25, 28]. The problems are mainly that students are not interested in learning history and have difficulty in memorizing too old and too many historical facts. Noraini [25] also mentioned that students have no interest in history learning because there is too much and too boring text contents. Additional problems are lack of teaching aid/material and not proficient in Malay Language.

Interviews with history teachers [12, 24, 25, 28, 30] also showed that there is currently no history courseware used in schools. They agreed that game based learning can help by providing interactive elements for visualizing the continuity of historical events. They also supported the development of history courseware using GBL approach to increase

students' interest and performance in history [28] and to reduce students' boredom with history learning [30].

Table 4.1. Problems in history learning from students' perception.

Problems in history learning	Percentage (%)
Have no interest in history because of teaching media such as boring text books.	44.5
Have no interest in history because of teacher's boring teaching method.	15.8
Lack of teaching aid/material used by the teachers.	16.5
Lack of history references.	19.6
Did not get clear descriptions about the historical events.	38.8
Difficult to memorize historical facts.	70.6
Did not understand history context.	37.3
Other.	7.9

Table 4.2. Problems in history learning from teachers' perception.

Students' weaknesses in history learning	Percentage (%)
Did not get clear descriptions about the historical events.	66.7
Difficult to memorize historical facts.	100
Did not understand history context.	60
Have no interest in learning history.	100
Lack of history reference.	0
Other.	26.7

Findings from this survey also indicated that, 92% students have experienced playing digital games. 27.7% of them played games for less than 1 hour per week while 16.4% played for 1 hour per week. Table 4.3 shows students' frequency of playing games.

Table 4.3. Frequency of playing digital/electronic games.

Frequency	Percentage (%)
< 1 hour/ week	27.7
1 hour/ week	16.4
2 hours/ week	15.8
3 hours/ week	13.6
4 hours/ week	5.6
> 4 hours/ week	13.6

Different platforms for digital/electronic games are television based system (such as the Sony PlayStation® series, Nintendo GameCube™, Microsoft's Xbox®, Xbox 360™), Computer/PC-based DVD-ROM or CD-ROM, Games-specific handheld consoles (such as Nintendo's Game Boy® and DS™ series, Sony's PSP™) using button and joystick control and other handheld unit [mobile computer, mobile phones, Personal Digital Assistants

(PDA)]. Table 4.4 shows the percentage of students using different platforms for playing games, 66.7% use other handheld unit. Table 4.5 shows the percentage of students playing different types of digital/electronic games. Most popular genres are racing games (66.7%) and adventure games (62.1%).

Table 4.4. Percentage of students using different platforms for playing digital/electronic games.

Forms of digital/electronic games	Percentage (%)
Television based system (Sony PlayStation® series, Nintendo GameCube™, Microsoft's Xbox®, Xbox 360™)	61.6
PC-based DVD ROM or CD ROM	58.8
Games-specific handheld consoles (Nintendo's Game Boy® and DS™ series, Sony's PSP™) with using button and joystick control)	41.8
Other handheld unit [mobile computer, mobile phones, P.D.As (Personal Digital Assistants)]	66.7

Table 4.5. Percentage of the students playing different types of digital/electronic games.

Types of digital/electronic games	Percentage (%)
Tactical shooters	42.4
Role playing	18.6
Racing	67.2
Car combat	26.6
sports game	50.8
adventure games	62.1
Fighting games	55.9
Combat Sims	19.8
Puzzle games	42.4
Rhythm games	21.5
Other	4.0

Furthermore, 83.6% of the students stated that they like to play digital/electronic games, one reason being gaming is fun, while 62.7% play games just to fill up their free time. Table 4.6 shows the percentage of students and their reasons for playing games. Table 4.7 presents the advantages of playing digital/electronic games.

Table 4.6. Percentage of students and their reasons for playing digital/electronic games.

Reasons why I like to play digital/electronic games	Percentage (%)
Fun.	72.9
Fill up free time.	62.7
Enthusiastic with adventure activity.	29.9
To show ability.	30.5
Enthusiastic with fantasy world.	45.2
Other.	6.8

Table 4.7. Percentage of students rating the advantages of playing digital/electronic games.

Advantages of playing digital/electronic games	Percentage (%)
Fun.	53.7
Improve electronic gaming skills.	39.0
Calming the thought.	49.2
A method to fill up the free time.	50.8
Increase creativity.	47.5
Other.	5.1

In conclusion, many students surveyed have played digital games previously, mostly using handheld units such as mobile phone and PDA. Most of them played racing and adventure games. 83.6% students like to play games and 69.5% students stated that playing digital games bring advantages to them. Findings from this preliminary analysis showed that the main requirements of game based history courseware are: illustrate history facts clearly, increase students' interest to learn history and learn history through creative experience. Table 4.8 shows the requirements of GBL history courseware from students' perception.

Table 4.8. Requirements of a GBL courseware from students' perception.

Requirements	Percentage (%)
Animate historical personality in a virtual environment to enable users to visualize the truth through role playing.	61.6
Animated history event in a virtual environment to enable users to visualize the truth through role playing.	54.8
Challenging adventure element.	53.1
Audio element for attraction to history learning.	52.5
Video element for attraction to history learning.	58.8
2D animation for attraction to history learning.	22.6
3D animation for attraction to history learning.	58.2
Graphic element for attraction to history learning.	49.7
Adaptive learning based on individuals' ability.	48.0
Increase students' interest in history learning.	66.7
Learn history via an entertained experience.	65.5
Illustrate historical facts clearly.	70.6
Other.	1.7

In summary, key findings from the preliminary analysis are: (1) types of problems faced by the students in learning history; (2) need for interesting,

creative teaching approach such as interactive history courseware to encourage history learning; (3) GBL approach has potential to be used in the development of interactive courseware to increase students' interest in learning history via an entertained experience; (4) 3D and multimedia technology should be used in development of interactive history courseware to illustrate historical facts clearly thus increase history learning.

4.3 GBL approach to teaching and learning of history

New electronic technologies have brought a revolution in the education system [33]. A growing number of researchers are exploring the potential of digital games for engaging students in learning experience [35]. One intriguing approach to engage students in studying world history is through computer simulation games *Civilization III* which is an entertainment game with deep historic, geographic and political simulation [35]. A museum online game EVE on the other hand, is based upon the richness of content from collection objects to provide a deeper level of intellectual stimulation and history learning. In order to engage students in learning more about museum object, online educational games become a popular option to extend the leaning experience [9].

We believe that GBL approach has the potential to help overcome problems in history learning, especially for the younger IT generation students.

4.4 Proposed components in GBL Model for history courseware design

Using ICT is important in teaching and learning of history because ICT can stimulate thinking skills and develop history thinking skills besides being a tool to get relevant resources [31]. Multimedia can make teaching and learning history easier and more interesting. History components which are designed by combining real photo concept and interesting animation can produce 'a live' performance [15]. Investigations of the features and design component for history courseware using GBL approach are essential to produce a courseware that can sustain learners' interest and to achieve learning goals. Figure 4.1 shows the proposed components in the form of a Model for history courseware design using GBL approach. The main components identified are pedagogy and digital games. For the pedagogy component, several elements are elaborated as follows:

(A) Learning goal setting

A clear learning goal setting in history courseware design with GBL is important to help accomplish the learning goals for history learning.

(B) Learning theory setting

Selection of suitable learning theories for history learning is important to foster the learning process effectively.

(C) Educational Psychology

Educational psychology is a discipline concerned with the overall teaching and teaching process, for example, the influence of development and differences among individuals on the process of learning [16]. Therefore, educational psychology is one of the crucial components in history courseware design for delivering an effective history lesson to learners.

(D) Country curriculum needs

Each country has different curriculum. Integrated Curriculum for Secondary School (KBSM) is a curriculum that implemented in Malaysian secondary schools. Therefore, KBSM components need to be considered for GBL approach to history courseware design to ensure that the courseware meets the curriculum needs.

(E) Patriotism and moral value

Patriotism and moral value that embedded in history learning through a history courseware with GBL approach is important to foster the spirit of loyalty to the country.

(F) Memorization & forgetting theory

According to Davidovitch et al. [7], the learning process contains forgetting components, which is applicable when the learning phase is ended or when the learning process includes breaks. A study on the process of learner memorization and how forgetting occurred especially in history learning, is important for designing a history courseware which can help the student to memorize history more effectively.

For Digital Games component, several components that need to be considered during design include:

(A) Game story's background

Story background of every game is important and it shows how history events happened. Features of each history events which consist of date, location and time need to be designed properly and not digressed from original history facts which stated in history syllabus.

(B) Rules

Clear instructions and rules for every game are very important. It is a guide line and regulations for learner to follow when they involved in the game play.

(C) Immersive

The game design should make learners feel immersive and absorbed when they involved in each game play.

(D) Enjoyment

Enjoyment is a crucial component that makes the learners enjoy with the game play. Without enjoyment of game play, the learners will simply give up of the game play.

(E) Feed back

The learners feed back is the learning response in the game instantly. The user interface design of the game design should make it easy for learners to give feed back to the game.

(F) Multimedia technology

Multimedia effects with 2D or 3D animations, immersive graphical environment and imitative sound are important to attract the learners to involve themselves in every game play.

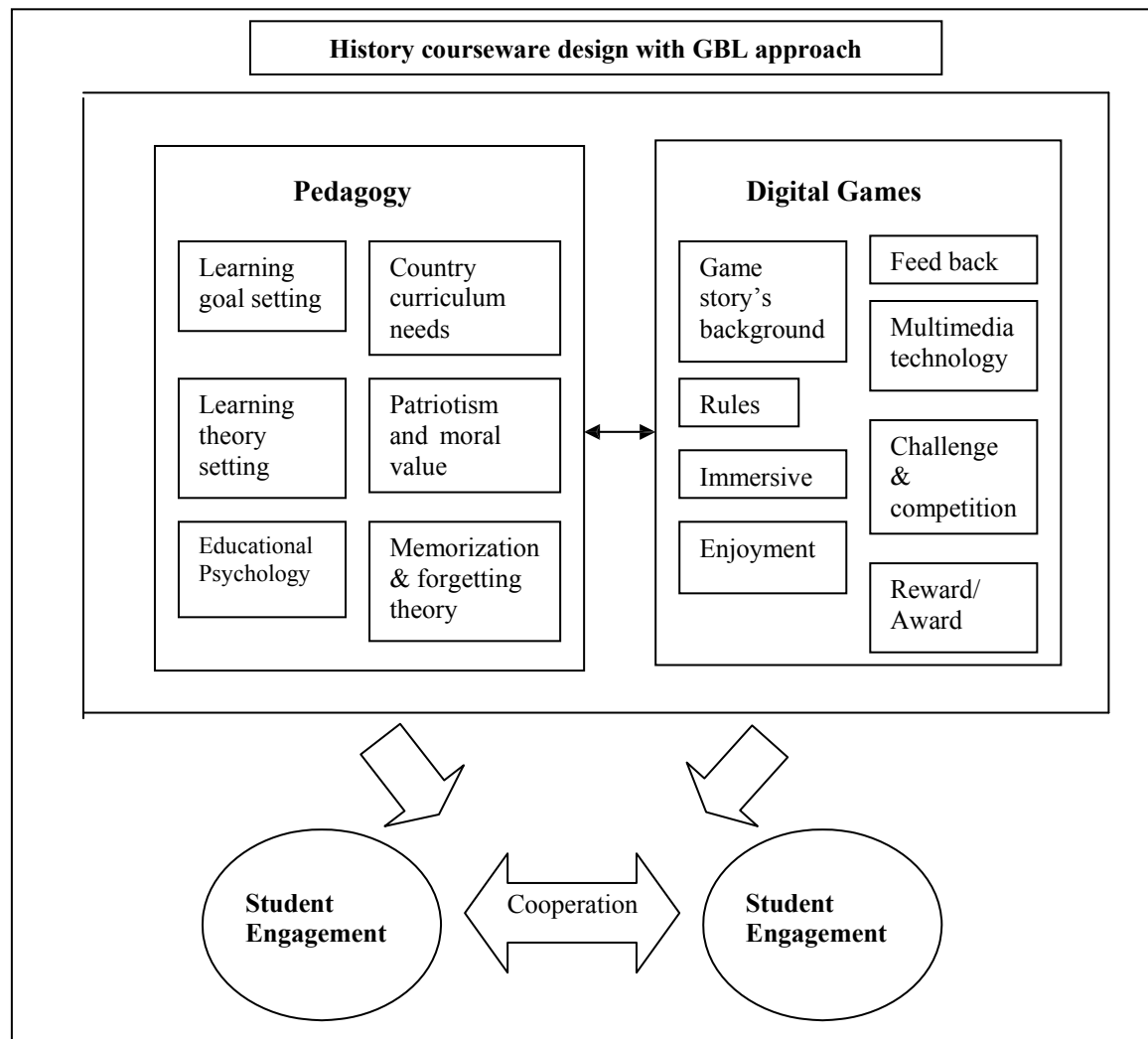


Figure 4.1. Proposed components in GBL Model for history courseware design.

(G) Challenge & competition

Every student has different abilities for every game play. Challenge of one game play should match the learner skill level without being boring and give up easily. Competition setting for each history game design is important for game level and so is the difficulties setting.

(H) Reward/Award

Reward/award is the one of the way to motivate the learners continues the game play and overcome the difficulties of each game level.

Student engagement and cooperation among students in every game play is the way of giving mutual support in history learning together towards accomplishing the learning goal.

5. Conclusions

In this paper, we reported findings from preliminary analysis of problems in history learning. The main problem of history learning is that students and teachers perceived history as a boring subject and it is difficult to memorized facts. Digital games, which most students enjoy playing can be used as an alternative approach to an otherwise boring subject. Therefore, based on problems identified, we proposed the use of game-based history learning courseware. A model for history courseware design with GBL approach is presented in this paper.

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